— *Questions marked with a star* ⋆ *are intended to be more difficult and may be skipped* —

1. Without computing any integrals, find the (complex) Fourier series of

$$\text{(i) } f(t) = \cos^3(2t), \quad \text{(ii) } g(t) = \cos(2t)\sin(4t),$$
$$\text{(iii) } h(t) = \sin(t)\cos(2t) + \sin(2t)\cos(t),$$

   Using these results, find the trigonometric Fourier series of the above functions.

   Hint. Recall that $\cos x = \frac{1}{2}(\mathrm{e}^{\mathrm{i}x} + \mathrm{e}^{-\mathrm{i}x})$ and $\sin x = \frac{1}{2\mathrm{i}}(\mathrm{e}^{\mathrm{i}x} - \mathrm{e}^{-\mathrm{i}x})$.

2. Find the complex Fourier series of the 2-periodic function defined on $[0, 2)$ by

$$f(t) = \begin{cases} t, & 0 \le t \le 1, \\ 1, & 1 < t < 2. \end{cases}$$

3. Let $f : \mathbb{R} \to \mathbb{C}$ be the 2π-periodic function defined by $f(t) = \mathrm{e}^{|t|}$ on the interval $[-\pi, \pi)$. Show that the Fourier series of f is given by

$$f(t) \sim \frac{1}{\pi} \sum_{n=-\infty}^{\infty} \frac{(-1)^n \mathrm{e}^{\pi} - 1}{n^2 + 1} \mathrm{e}^{\mathrm{i}nt}.$$

   Without computing any further integrals, compute the trigonometric form of the Fourier series.

4. Plot the even and odd extensions of the following functions:

$$\begin{aligned} &\text{(i)} && f(t) = (t-1)^2, && 0 < t < \pi, \\ &\text{(ii)} && g(t) = \mathrm{e}^t - 1, && 0 < t < 1, \\ &\text{(iii)} && h(t) = \frac{1}{1+t}, && 0 < t < 2. \end{aligned}$$

   In each case indicate where the function meets the coordinate axes. Find the cosine half-range series of $f(t)$.

5. Let $f : [0, 2) \to \mathbb{R}$ be the function

$$f(t) = \begin{cases} t, & 0 \le t < 1, \\ 2 - t, & 1 \le t < 2. \end{cases}$$

   Show that the sine half-range series of f is given by

$$f(t) \sim \sum_{n=1}^{\infty} \frac{8}{\pi^2 n^2} \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{\pi n t}{2}\right).$$

6. (⋆) Let $f : \mathbb{R} \to \mathbb{R}$ be the function defined by $f(t) = \sqrt{t}$ on $[0, \pi)$. Show that the sine half-range series can be written as

$$f(t) \sim \sum_{n=1}^{\infty} \left(\frac{2(-1)^{n+1}}{n\sqrt{\pi}} + \frac{2C(\sqrt{\pi n})}{\pi n^{3/2}} \right) \sin nt,$$

where $C(x)$ is the Fresnel integral

$$C(x) = \int_0^x \cos(t^2)\, \mathrm{d}t.$$

7. Show that the function $f(t)$ given in problem 2 satisfies the Dirichlet criterion. By evaluating the Fourier series of f at $t = 0$ and equating it to the appropriate value, derive that

$$\sum_{n=1}^{\infty} \frac{1 + (-1)^{n+1}}{n^2} = \frac{\pi^2}{4}.$$

8. By applying Parseval's theorem to the sine half-range series for the function $f(t)$ in problem 5, show that

$$\sum_{k=1}^{\infty} \frac{1}{(2k-1)^4} = \frac{\pi^4}{96}.$$

Hint. Remember that the sine half-range series is the Fourier series of the odd extension F_{odd} of f.